WORKED SOLUTION

STEP II 2016, Question 12

Probability and Statistics • Inclusion-Exclusion and Derangements

STEP II 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 9.8, *Nobody Gets Their Own Shirt*: https://beyondstep.keithmcnulty.org/ch09/derangements.html

The key idea

For three events, treat $A \cup B$ as a single event and apply the two-event rule twice. The same alternating pattern (add singles, subtract pairs, add triples, ...) holds for any number of events. Applied to the events "card i is in place i", every term with r events has the same probability, so each layer of the sum collapses to a single fraction. For "exactly one", fix which card is the match and ask for the rest to have no match at all.

Solution

Three events

Write $A \cup B \cup C = (A \cup B) \cup C$ and use the two-event rule with $A \cup B$ in place of A and C in place of B:

$$\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A \cup B) + \mathrm{P}(C) - \mathrm{P}\big((A \cup B) \cap C\big).$$

The first term is $\mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$. For the last term, the distributive law gives $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$, and applying the two-event rule once more,

$$\begin{aligned}\mathrm{P}\big((A \cap C) \cup (B \cap C)\big) &= \mathrm{P}(A \cap C) + \mathrm{P}(B \cap C) - \mathrm{P}\big((A \cap C) \cap (B \cap C)\big) \\ &= \mathrm{P}(A \cap C) + \mathrm{P}(B \cap C) - \mathrm{P}(A \cap B \cap C).\end{aligned}$$

Substituting both,

$$\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(A \cap B) - \mathrm{P}(B \cap C) - \mathrm{P}(C \cap A) + \mathrm{P}(A \cap B \cap C),$$

as required.

Four events

Add the four single probabilities, subtract the six pairwise intersections, add the four triple intersections and subtract the quadruple one:

$$\begin{aligned}\mathrm{P}(A\cup B\cup C\cup D) = {} & \mathrm{P}(A)+\mathrm{P}(B)+\mathrm{P}(C)+\mathrm{P}(D)\\ & -\mathrm{P}(A\cap B)-\mathrm{P}(A\cap C)-\mathrm{P}(A\cap D)\\ & -\mathrm{P}(B\cap C)-\mathrm{P}(B\cap D)-\mathrm{P}(C\cap D)\\ & +\mathrm{P}(A\cap B\cap C)+\mathrm{P}(A\cap B\cap D)+\mathrm{P}(A\cap C\cap D)+\mathrm{P}(B\cap C\cap D)\\ & -\mathrm{P}(A\cap B\cap C\cap D).\end{aligned}$$

Parts (i)–(iii)

All $n!$ orderings are equally likely. If r particular cards are required to sit in their own positions, the remaining $n-r$ cards can be arranged in $(n-r)!$ ways, so the probability is $(n-r)!/n!$. Hence

$$\text{(i) } \mathrm{P}(E_i)=\frac{1}{n}, \qquad \text{(ii) } \mathrm{P}(E_i\cap E_j)=\frac{1}{n(n-1)}, \qquad \text{(iii) } \mathrm{P}(E_i\cap E_j\cap E_k)=\frac{1}{n(n-1)(n-2)}.$$

(Equivalently, for (ii): card i lands in place i with probability $\frac{1}{n}$, and then card j lands in place j among the $n-1$ places left with probability $\frac{1}{n-1}$.)

At least one match

At least one card matches exactly when $E_1\cup E_2\cup\cdots\cup E_n$ occurs. The pattern above extends to n events: the rth layer is $(-1)^{r+1}$ times the sum of the probabilities of all intersections of r distinct events.

> **Ask yourself** How can a sum over all choices of r events be evaluated without knowing which events they are?

By symmetry every intersection of r distinct events has the same probability, $(n-r)!/n!$, and there are $\binom{n}{r}$ ways to choose the r events. So the rth layer contributes

$$(-1)^{r+1}\binom{n}{r}\frac{(n-r)!}{n!}=(-1)^{r+1}\frac{n!}{r!\,(n-r)!}\cdot\frac{(n-r)!}{n!}=\frac{(-1)^{r+1}}{r!}.$$

Summing over $r=1,2,\ldots,n$,

$$\mathrm{P}(\text{at least one match})=1-\frac{1}{2!}+\frac{1}{3!}-\cdots+(-1)^{n+1}\frac{1}{n!},$$

as required. (For $r=1,2,3$ this is just $n\cdot\frac{1}{n}$, $\binom{n}{2}\cdot\frac{1}{n(n-1)}$ and $\binom{n}{3}\cdot\frac{1}{n(n-1)(n-2)}$ from parts (i)–(iii).)

Exactly one match

Write p_m for the probability that none of m shuffled cards is in its own place. By the previous result,

$$p_m = 1 - \left(1 - \frac{1}{2!} + \cdots + (-1)^{m+1}\frac{1}{m!}\right) = \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^m \frac{1}{m!}.$$

The event "exactly one match" is the disjoint union over i of the events "card i matches and no other card does". Given E_i, the other $n-1$ cards are equally likely to be in any order in the other $n-1$ positions, so this is a fresh shuffle of $n-1$ cards, and the probability that none of them matches is p_{n-1}. Therefore

$$\mathrm{P}(\text{exactly one match}) = \sum_{i=1}^{n} \mathrm{P}(E_i)\, p_{n-1} = n \cdot \frac{1}{n} \cdot p_{n-1} = p_{n-1},$$

that is,

$$\mathrm{P}(\text{exactly one match}) = \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^{n-1}\frac{1}{(n-1)!} \qquad (n \geq 2),$$

and 1 when $n = 1$. Equivalently, it is $\sum_{r=0}^{n-1}(-1)^r/r!$, which covers $n = 1$ too.

Checks

For $n = 3$ the orderings are 123, 132, 213, 231, 312, 321. Four of them have a match, giving $\frac{4}{6} = \frac{2}{3} = 1 - \frac{1}{2} + \frac{1}{6}$; and three (the swaps 132, 213, 321) have exactly one match, giving $\frac{1}{2}$, which agrees with the formula. For $n = 2$, exactly one match is impossible (if one card is in place, so is the other), and the formula gives 0. A computer enumeration of all orderings for $n \leq 7$ agrees with both formulas; for example, for $n = 7$ exactly one match has probability $\frac{53}{144} \approx 0.368$. As $n \to \infty$ both answers tend to familiar limits: at least one match tends to $1 - \mathrm{e}^{-1} \approx 0.632$, and exactly one match tends to e^{-1}.

Common slips

- Writing $\mathrm{P}(E_i \cap E_j) = \frac{1}{n^2}$: the events are not independent, because once card i is in place i only $n-1$ places remain for card j.
- Answering "exactly one" with $n \cdot \frac{1}{n} = 1$ or with $\mathrm{P}(E_1)$: we need card i in place *and* no other card in place.
- Forgetting that the term in the sum for "at least one" with $r = 1$ is $n \cdot \frac{1}{n} = 1$, which is why the series starts with 1.

Beyond STEP

This is the classical *derangement* problem (the *problème des rencontres*), and the general inclusion–exclusion formula is proved neatly by induction or with indicator random

variables. The number of matches turns out to be approximately Poisson with mean 1 for large n: the probability of exactly k matches tends to $\mathrm{e}^{-1}/k!$. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both treat inclusion–exclusion and matching problems.

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